

“Not Every Institute Can Or Should Set Up A Semiconductor Lab”

What kinds of jobs are emerging from the massive investments in India's semiconductor industry? Do we have the talent to meet this demand? If not, how are the government, academia, and industry addressing the gap—and how can these efforts be accelerated to build the right talent faster? To explore these questions, EFY's Rahul Chopra had the opportunity to speak with Dr Ashwini K Aggarwal, a seasoned leader who has worn many hats within the semiconductor ecosystem. Here is what he shared.

DR ASHWINI K AGGARWAL

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What kind of job opportunities or career opportunities are emerging now because of the growth in the semiconductor sector and all the investments happening?

The semiconductor industry in India is reaching an inflection point. With major mega and giga projects announced, we are seeing investments of close to ₹150 billion being implemented. Tata's fab is progressing at a great pace. Projects from companies such as Kaynes, Micron, CG Semi, and many smaller players are also taking shape.

As this unfolds, it is inevitable that India will need to develop

manufacturing skills. While India has long been designing chips, we now need to design our future in the 'Make in India' era. Careers are evolving, and we are seeing growing demand for skilled professionals across the entire ecosystem, not just at fabs but throughout the value chain.

Can you give some examples of the kinds of jobs being created?

Let me quote the CEO of Applied Materials. In a meeting, he once said, "I wish I had people who knew atomic layer deposition or could understand plasma engineering in the country; I would give 10

job letters on the spot."

Design remains key, no doubt. But we are now completing the circle with manufacturing. A fab, as a process industry, requires more than electronics engineers—it also needs mechanical, materials, chemical, computer, electrical, and EHS (environment, health, and safety) engineers.

Each of these disciplines plays a crucial role in enabling the successful operation of a fab. In addition to technical roles, there is a significant demand for professionals in health and safety. Specialists who manage hazardous chemicals and